CHAPTER I

Projector Resolved Seed Calculus I

Typed operator systems, seed rank, and exact packet coordinates

The opening chapter introduces the seed and operator language used throughout the book.

Seeds, operators and projectors provide the linear language of the series. Exact packet coordinates make the generated objects and the maps between them visible.

IN THIS CHAPTER

Mathematical text 2

Formal companion 1588

Sources and references 2666

Scope

The correspondence concerns the stated proof-level constructions. Noncomputable existence results do not provide executable solvers.

The next chapter compares these descriptions across refinement levels and asks which structures survive that passage.

CHAPTER II

Projector Resolved Seed Calculus II

Refinement towers, weighted transfer, and integral separation of operator packets

The linear framework now becomes a language for comparing resolutions.

Refinement towers compare packet descriptions across levels. Weighted transfers and integral separation show which structures survive a change of resolution.

IN THIS CHAPTER

Mathematical text 99

Visual studies 152

Formal companion 1721

Sources and references 2668

Scope

Canonical values are defined for admissible data. A general existence theorem for admissible contours remains a separate question.

The next chapter moves from linear packet coordinates to typed nonlinear operations and analytic jets.

CHAPTER III

Projector Resolved Seed Calculus III

Typed transfer, nonlinear packets, and obstruction theory

The preceding constructions provide the linear setting. This chapter studies nonlinear coordinates and the limits of their extension.

Typed operations extend the linear language to nonlinear coordinates, analytic packet jets and obstruction theory. The analytic supplement develops the continuous packet-jet claims.

IN THIS CHAPTER

Mathematical text 178

Analytic supplement 248

Visual studies 254

Formal companion 1768

Sources and references 2670

Scope

Expanded ordinary proofs and the bounded Lean core have different coverage. The packet Taylor estimates, counting arguments and later extensions retain their stated ordinary-proof status. Arbitrary differentiability is not asserted.

The next chapter develops a cyclic analytic family through coefficient selection, transport and spectral data.

CHAPTER IV

Projector Resolved Seed Calculus IV

Cyclic analytic functions, reciprocal families, and spectral zero data

The focus shifts from the general packet language to a particular analytic family.

A cyclic analytic family connects coefficient selection, differential transport, reciprocal polynomials and spectral zero data.

IN THIS CHAPTER

Mathematical text 280

Visual studies 355

Formal companion 1815

Sources and references 2672

Scope

The analytic statements retain their branch choices, multiplicities and integer-order hypotheses. Universal zero simplicity and a zeta functional equation remain outside these results.

The next chapter turns to signals, where filters, packet trees and reconstruction make changes of resolution concrete.

CHAPTER V

Multiresolution Signal Calculus

Exact coordinates, singular packets, and stable reconstruction

The analytic constructions give way to signal representations and reconstruction.

Signal models move through filters, packet trees and reconstruction maps. Exact coordinates and singular packets distinguish representation from approximation and recovery.

IN THIS CHAPTER

Mathematical text 383

Visual studies 467

Formal companion 1936

Sources and references 2675

Scope

Separated-jump results require the stated model, gap and grid assumptions. Stable edge-packet recovery and the broader approximation class remain open.

The next chapter asks how transfers evolve in time and how eliminated states contribute memory.

CHAPTER VI

Transfer Semigroups and Memory

Probabilistic refinement, fractional operators, and exact hidden-state elimination

Static descriptions now become evolution and conditioning problems.

Evolution, probability and conditioning share a transfer language. Eliminating hidden states produces memory terms whose hypotheses remain part of the model.

IN THIS CHAPTER

Mathematical text 511

Visual studies 616

Formal companion 1992

Sources and references 2678

Scope

Finite-state, common-gap and fixed-partition restrictions remain explicit. Completion, uniform refinement control and stable infinite latent-packet closure remain open. Floating residuals are not interval certificates.

The next chapter treats quantum seeds, operator grammars and finite effective dynamics.

CHAPTER VII

Quantum Seeds and Effective Dynamics

Symmetry, geometric transport, quantum channels, and bounded refinement towers

The transfer viewpoint is now developed in quantum settings with their own algebraic and dynamical assumptions.

Quantum seeds generate spaces through an operator grammar. Symmetry, circuits, geometric transport and quantum channels lead to finite effective dynamics and bounded refinement questions.

IN THIS CHAPTER

Mathematical text 660

Visual studies 772

Formal companion 2054

Sources and references 2681

Scope

The finite formal contract does not certify the full continuum physics program. Anomaly-free refinement, uniform Feshbach control, Lorentzian reconstruction and the proposed gravitational limits remain open.

The next chapter turns to algorithms, separating exact structural reductions from numerical and hardware questions.

CHAPTER VIII

Packet Operator Algorithms

Exact compilation and GPU realization

The earlier packet constructions become inputs to exact compilation and numerical implementation.

Structured packet representations remove work before execution. Exact compilation connects algebraic structure, recompression, precision and stated hardware observations.

IN THIS CHAPTER

Mathematical text 814

Visual studies 897

Formal companion 2132

Sources and references 2684

Scope

Small-core optimality uses supplied ordered thin-SVD equations and actual compact ranges. General numerical SVD verification, unrestricted discovery and CUDA-kernel verification require further results.

The book then turns to arithmetic, beginning with Euler congruences, completion and dyadic geometry.

CHAPTER IX

Euler Congruences

Completion, factorial strengthening, and sharp dyadic geometry

The arithmetic sequence begins with congruence identities and the hypotheses needed to extend them.

Euler congruences are developed through completion, factorial strengthening and dyadic geometry. The accompanying guides connect exact statements, bounded examples and formal coverage.

IN THIS CHAPTER

Mathematical text 931

Visual studies 1145

Formal companion 2220

Sources and references 2687

Scope

The arithmetic, dyadic and lifting results have distinct hypotheses. General Euler-tail simplicity and the broader native and polyphase extensions remain open.

The next chapter organizes residual seeds and conditional system completion from supplied arithmetic data.

CHAPTER X

Universal Residual Seeds

Deformation forests, transport categories, and conditional system completion

The congruence study is followed by a framework for residual seeds and compatible arithmetic descriptions.

Residual seeds, deformation forests and transport categories organize conditional system completion. Supplied arithmetic and carrier data remain explicit inputs.

IN THIS CHAPTER

Mathematical text 1170

Visual studies 1227

Formal companion 2350

Sources and references 2690

Scope

The target family and recovered carrier are independent existential outputs. Identifying them and proving nonzero occupancy require additional premises. Source families, carrier presentations and local comparisons remain supplied.

The next chapter studies local packet geometry and the remaining obligations of arithmetic realization.

CHAPTER XI

Tame and Wild Packet Geometry

For local Langlands

The preceding chapter separates the arithmetic objects supplied as inputs from those obtained conditionally. Local packet geometry sharpens the realization questions.

Tame and wild packet geometry organizes the local arithmetic constructions. The research companions retain both restricted progress and the unresolved obligations of Conjecture 66.

IN THIS CHAPTER

Mathematical text 1260

Visual studies 1367

Formal companion 2384

Further arithmetic studies 1482

Sources and references 2693

Scope

Conjecture 66 remains open. Its unresolved requirements concern general arithmetic realization, H5 for the same candidate, corrected full-character occurrence and actual native data under the stronger contract.

The final chapter gives a conditional route through patching support and Hecke compatibility. The open realization obligations remain separate.

CHAPTER XII

Patching Support and Hecke Packets

Conditional automorphic completion

The arithmetic sequence culminates in a conditional argument using patching support, Hecke compatibility and the stated representation-theoretic inputs.

Patching support and compatible Hecke packets complete a conditional arithmetic argument. Each application must supply the geometric, automorphic and representation-theoretic inputs that the theorem uses.

IN THIS CHAPTER

Mathematical text 1418

Visual studies 1455

Formal companion 2617

Sources and references 2702

Scope

Actual residual and deformation data, applicable patching and full support, a compatible same-field Hecke point, classicality and an attached representation are required. Comparisons of compatible families assume those families already exist. Chapter XI's Conjecture 66 is not assumed proved.

The formal companions at the back of the book describe their own precise coverage. The glossary provides a route back to the principal definitions.

Further studies

The following studies develop the restricted arithmetic constructions, obstructions and remaining realization questions associated with Chapter XI.

READING ROUTES

Arithmetic realization and its obstructions 1483

Visual studies of arithmetic realization 1551

FURTHER STUDY

Arithmetic realization and its obstructions

These studies distinguish finite packet constructions, actual arithmetic candidates, full-character occurrence and comparison for the same candidate.

READING ROUTES

Research companion 1484

Scope

Conjecture 66 remains open. Its unresolved requirements concern general arithmetic realization, H5 for the same candidate, corrected full-character occurrence and actual native data under the stronger contract.

FURTHER STUDY

Visual studies of arithmetic realization

The plates give worked constructions and counterexamples, with explicit links to the corresponding arguments and their hypotheses.

READING ROUTES

Research atlas 1552

Formal companions

The formal companions connect the mathematical chapters to their Lean declarations, hypotheses, source correspondences and verification evidence. Read each statement with the coverage described by its companion.

READING ROUTES

I Projector Resolved Seed Calculus I 1588

II Projector Resolved Seed Calculus II 1721

III Projector Resolved Seed Calculus III 1768

IV Projector Resolved Seed Calculus IV 1815

V Multiresolution Signal Calculus 1936

VI Transfer Semigroups and Memory 1992

VII Quantum Seeds and Effective Dynamics 2054

VIII Packet Operator Algorithms 2132

IX Euler Congruences 2220

X Universal Residual Seeds 2350

XI Tame and Wild Packet Geometry 2384

XII Patching Support and Hecke Packets 2617

FORMAL COMPANION I

Projector Resolved Seed Calculus I

The formal companion to Chapter I explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Computational formulation 1589

Formal companion 1683

Return to Chapter I

FORMAL COMPANION II

Projector Resolved Seed Calculus II

The formal companion to Chapter II explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 1722

Return to Chapter II

FORMAL COMPANION III

Projector Resolved Seed Calculus III

The formal companion to Chapter III explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 1769

Return to Chapter III

FORMAL COMPANION IV

Projector Resolved Seed Calculus IV

The formal companion to Chapter IV explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 1816

Return to Chapter IV

FORMAL COMPANION V

Multiresolution Signal Calculus

The formal companion to Chapter V explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 1937

Return to Chapter V

FORMAL COMPANION VI

Transfer Semigroups and Memory

The formal companion to Chapter VI explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 1993

Return to Chapter VI

FORMAL COMPANION VII

Quantum Seeds and Effective Dynamics

The formal companion to Chapter VII explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2055

Return to Chapter VII

FORMAL COMPANION VIII

Packet Operator Algorithms

The formal companion to Chapter VIII explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2133

Return to Chapter VIII

FORMAL COMPANION IX

Euler Congruences

The formal companion to Chapter IX explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2221

Return to Chapter IX

FORMAL COMPANION X

Universal Residual Seeds

The formal companion to Chapter X explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2351

Return to Chapter X

FORMAL COMPANION XI

Tame and Wild Packet Geometry

The formal companion to Chapter XI explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2385

Return to Chapter XI

FORMAL COMPANION XII

Patching Support and Hecke Packets

The formal companion to Chapter XII explains the declared hypotheses, theorem correspondences and evidence boundaries.

IN THIS COMPANION

Formal companion 2618

Return to Chapter XII

Glossary

Definitions and reading references for the recurring language of the twelve chapters. Local meanings and hypotheses remain those of the cited mathematical treatment.

BROWSE ALPHABETICALLY

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Sources and references

Sources are grouped by chapter and treatment. Citation labels retain the numbering or short names used in the text. The accompanying files contain the editable material and supporting source records.

BY CHAPTER AND TREATMENT

Chapter I mathematical text 2666

Chapter II mathematical text 2668

Chapter III mathematical text 2670

Chapter IV mathematical text 2672

Chapter V mathematical text 2675

Chapter VI mathematical text 2678

Chapter VII mathematical text 2681

Chapter VIII mathematical text 2684

Chapter IX mathematical text 2687

Chapter X mathematical text 2690

Chapter XI mathematical text 2693

Arithmetic realization studies: direct construction 2695

Arithmetic realization studies: monomial routes and Clifford descent 2698

Arithmetic realization studies: conductors and native markings 2700

Chapter XII mathematical text 2702

Chapter I mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2667

Return to Chapter I

Chapter II mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2669

Return to Chapter II

Chapter III mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2671

Return to Chapter III

Chapter IV mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2673

Return to Chapter IV

Chapter V mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2676

Return to Chapter V

Chapter VI mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2679

Return to Chapter VI

Chapter VII mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2682

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Chapter VIII mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2685

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Chapter IX mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2688

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Chapter X mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2691

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Chapter XI mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2694

Return to Chapter XI

Arithmetic realization studies: direct construction

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Research companion 2696

Return to the arithmetic studies

Arithmetic realization studies: monomial routes and Clifford descent

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Research companion 2699

Return to the arithmetic studies

Arithmetic realization studies: conductors and native markings

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Research companion 2701

Return to the arithmetic studies

Chapter XII mathematical text

The citation labels below belong to this treatment. References in the mathematical text link to the corresponding entries.

READING ROUTES

Mathematical text 2703

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